

# Vlad Voropaev

Computer Vision Engineer | Industrial Video Analytics | Detection, Tracking & Pose Estimation

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## SUMMARY

Computer Vision Engineer with 4+ years of experience building industrial video analytics and applied machine learning systems. Built and iterated on multi-camera safety systems from dataset design and held-out evaluation through object tracking, pose-based event logic, RTSP integration, field testing, and paid client customization. First author and oral presenter of an IEEE ACDSA 2024 paper.

## TECHNICAL SKILLS

**Computer Vision:** Object detection, object tracking, multi-camera video analytics, pose estimation, PPE and zone analytics, OCR, image preprocessing

**ML & Libraries:** Python, PyTorch, OpenCV, Ultralytics YOLO, TensorFlow, ByteTrack, Deep SORT, NumPy

**Systems & Evaluation:** RTSP, FastAPI, Docker, PostgreSQL, Linux, Git, GPU inference, dataset design and annotation, held-out evaluation, error analysis

## EXPERIENCE

**LLC Cyclop** - Co-founder, Deputy CEO & Applied AI Engineer (Computer Vision) | Oct 2022 - Jul 2026

*Industrial Computer Vision, client customization, pilot integration, product delivery, and R&D*

- Owned end-to-end engineering of a six-camera crane-operator safety system: built a **1,163-image dataset with 7,499 labeled objects**, trained YOLOv8m, and achieved **0.932 precision, 0.878 recall, 0.920 mAP@0.50, and 0.707 mAP@0.50:0.95** on a held-out test set; the work became a paid client customization
- Implemented active-view selection, ByteTrack identities, pose-based wrist-to-pendant association, helmet checks, temporal voting, and saved event evidence; replaced the initial IoU tracker and iterated after a white helmet failure observed during field testing
- Co-developed modular RTSP analytics for PPE, danger zones, tracking/counting, equipment use, and worker activity; offline PPE validation reached **0.906 mAP@0.50 and 0.772 mAP@0.50:0.95** on 579 images with 1,748 labeled objects
- For a multi-camera industrial pilot, manually reviewed **17,696 CCTV frames from eight camera feeds**. Curated **1,156 relevant frames** and designed view-specific zones to reduce long-range and duplicate detections across overlapping views

**Shtrih-M** - Computer Vision Engineer Intern | Feb 2022 - Jun 2022

- Developed and visually evaluated a Python/OpenCV preprocessing workflow across **1,147 raw GRBG Bayer images** captured under varied lighting, glare, backgrounds, and packaging conditions; combined demosaicing, highlight suppression, CLAHE, and adaptive gamma correction
- Documented seven image-processing functions, before/after comparisons, RGB histograms, adaptive decision logic, and implementation details in a **64-page technical report**

**Sberbank** - Data Scientist Intern, Research & Innovation | Sep 2021 - Nov 2021

- Developed and evaluated loan-delinquency modeling approaches using ROC-AUC; one model evaluation reached **0.89**

## RESEARCH & PUBLICATIONS

**NeuroQuest** - System Developer, First Author & Oral Presenter | 2023 - 2024

- Built an end-to-end RTSP-to-comic research system combining detection, tracking, face/action recognition, scene interpretation, language and image generation, and PDF assembly; led the paper's writing and presented the work at ACDSA 2024 ([IEEE Xplore 10467698](https://ieeexplore.ieee.org/abstract/document/10467698))

**First author:** "Automatic Generation of Neurocomics Based on Computer Vision System Data," IEEE ACDSA 2024 (DOI: [10.1109/ACDSA59508.2024.10467698](https://doi.org/10.1109/ACDSA59508.2024.10467698))

**Single author:** "Automatic Zone Detection in Images Using Polygonal Templates," Vestnik Nauki, 2023 (DOI: [10.24412/2712-8849-2023-1269-1067-1081](https://doi.org/10.24412/2712-8849-2023-1269-1067-1081))

## EDUCATION

**Bauman Moscow State Technical University** | Sep 2022 - Jul 2024

*M.S., Information Systems and Technologies, with honors*

**Bauman Moscow State Technical University** | Sep 2018 - Jul 2022

*B.S., Information Systems and Technologies, with honors*

## LANGUAGES

**Russian** - native; **English** - working proficiency